

Report

English and Scientific Methods Department
Critical Thinking and Scientific Method (SM101)

The relationship between video games and individuals' mental well-being

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I. Executive summary

The growing prevalence of video games as a primary leisure activity has raised concerns about their potential impact on individuals' mental well-being, particularly in terms of stress, motivation, sleep quality, and emotional regulation. The purpose of this report is to investigate the relationship between video game usage and depression indicators among individuals. **We hypothesize that playing video games causes depression.**

A questionnaire has been distributed online to 67 respondents via WhatsApp, LinkedIn, and Instagram. The questionnaire responses show that despite 20.9% of respondents spending more than three hours daily playing phone games, the majority reported only moderate depression indicators rather than severe ones. Furthermore, while 44.8% of respondents engage with digital entertainment daily, only 20.9% reported frequently experiencing trouble sleeping, and 26.9% reported usually or always feeling overwhelmed — levels that do not suggest a strong causal link between gaming and depression. Similarly, 26.9% frequently feel low in energy or unmotivated, yet this was observed across both high and low gaming groups alike.

To conclude, it has been found that video games are not the primary cause of depression indicators among respondents. The majority showed moderate rather than extreme stress-related behaviours, and most did not engage in prolonged daily gaming. **Based on the findings, the hypothesis that playing video games causes depression is rejected.** A further investigation into the effect of academic load on depression levels could be conducted. A suggested hypothesis for future investigation is: academic load causes depression among individuals.

II. Introduction

Video games have become one of the most widespread leisure activities globally, with millions of individuals across all age groups dedicating significant portions of their daily time to digital gaming and entertainment. While gaming is often regarded as a harmless recreational outlet, growing concerns have emerged regarding its potential impact on mental health, particularly in relation to stress, sleep quality, motivation, and emotional well-being. Studies suggest that spending excessive hours playing video games may reduce productivity, disrupt sleeping habits, and increase feelings of isolation and emotional strain (ChatGPT, OpenAI, 2026). On the other hand, some individuals consider gaming a means of relaxation and a way to decompress after demanding daily responsibilities, which may indicate a more complex relationship between gaming habits and psychological health (ChatGPT, OpenAI, 2026). Therefore, the relationship between video game usage and depression indicators remains an important topic to investigate across different individuals and lifestyles.

Thus, it is hypothesized that playing video games causes depression.

For the purpose of this report, video game usage is operationally defined as the daily amount of time an individual spends playing electronic or online games, including phone games, PC games, and digital entertainment platforms.

Meanwhile, depression is operationally defined as an individual's emotional and mental condition in terms of stress levels, energy, motivation, sleep quality, and ability to cope with daily pressures.

III. Body

A) Methodology

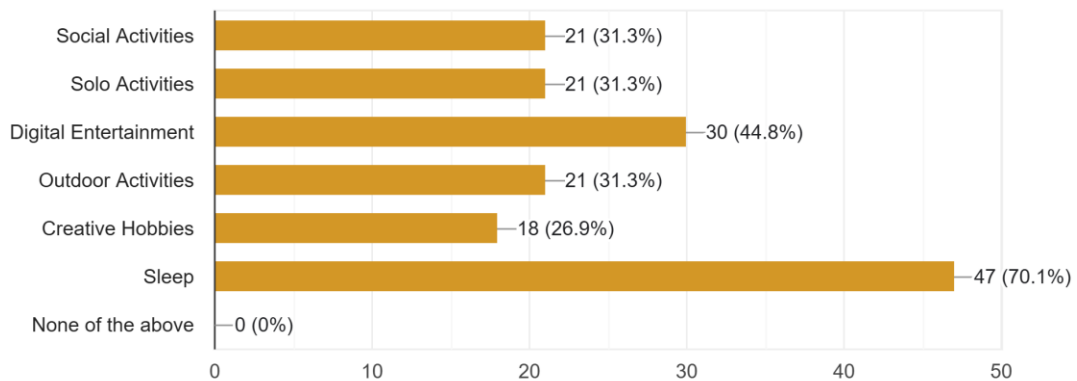
The survey has been distributed among 67 respondents, all of the 67 respondents are included; among the 67 participants there are 58 males and 9 females whose ages range between 14 and 45. The questionnaire consists of 8 questions, one Yes/No question, one check-list question, two multiple-choice questions, four scale question, and finally fill in the blanks questions (demographics). The questionnaire has been uploaded online via Google Forms, and the links have been dispersed via WhatsApp, LinkedIn and Instagram.

B) Findings

In the first question, participants were asked which activities they use to recharge after a long day. Since respondents could choose more than one option, the total exceeds 100%. The most frequently selected activity was Sleep, chosen by 70.1% of respondents. Digital Entertainment was chosen by 44.8% of participants, while Social Activities, Solo Activities, and Outdoor Activities were each selected by 31.3%. Creative Hobbies was the least chosen option, selected by 26.9% of respondents. These results suggest that gaming is one of several recharging strategies rather than the dominant or exclusive activity among participants.

1- Which of the following describes how you recharge after a long day? (Check all that apply)

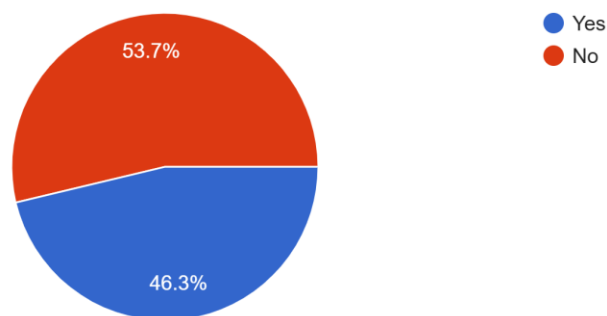
67 responses



In the second question, participants were asked whether they find it difficult to stay calm in stressful situations. A majority of 53.7% responded “No”, indicating that they generally manage to remain calm under stress, while 46.3% responded “Yes”. This indicates that emotional stress and anxiety-related reactions are relatively common among participants, which may be associated with symptoms of depression. However, the majority of respondents appear to cope adequately with stressful situations.

2- Do you find it difficult to stay calm in stressful situations? (Choose one answer)

67 responses

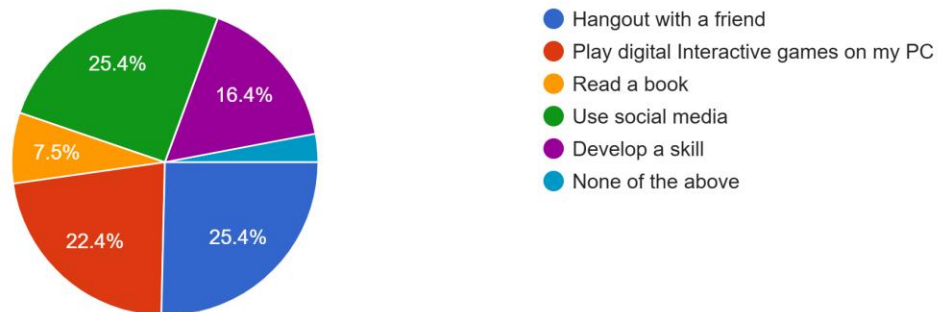


The third question asks about the most likely activity participants engage in during their free time. “Hanging out with a friend” and “Using social media” tied for the highest response, each selected by 25.4% of respondents. “Playing digital interactive games on a PC” was chosen by 22.4%, while

“Developing a skill” was selected by 16.4%. “Reading a book” accounted for 7.5% of responses, and 3% of participants chose “None of the above”. These findings indicate that digital gaming forms a significant part of participants’ free-time activities; however, it does not dominate. The presence of strong social activity also suggests that gaming does not necessarily lead to social isolation.

3- In your free time, what are you most likely to do? (Choose one answer)

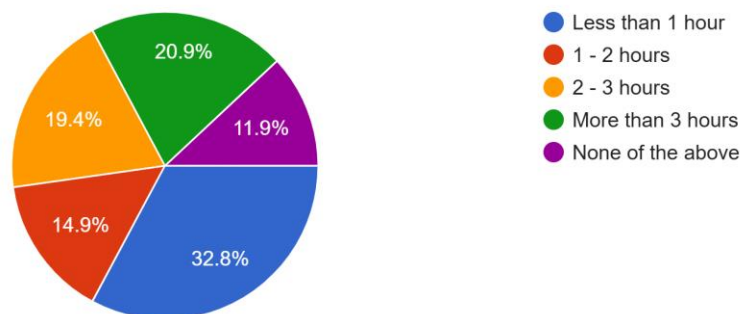
67 responses



The fourth question shows how much time participants spend gaming on their phone per day. The majority (32.8%) spend less than one hour, while 14.9% game between one and two hours per day, and 19.4% between two and three hours per day. A notable percentage of 20.9% spend more than three hours gaming on their phone daily. In addition, 11.9% reported not gaming on their phone at all.

4- On a typical day, how much time do you spend gaming on your phone? (Choose one answer)

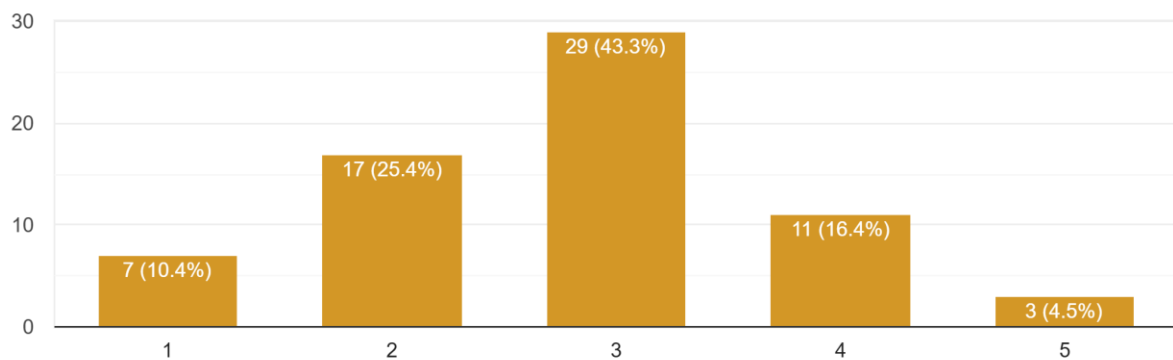
67 responses



In the fifth question, participants were asked how often they have trouble sleeping. The majority (43.3%) reported experiencing sleep difficulties sometimes, while 25.4% reported rarely experiencing them. A further 16.4% indicated that they usually have trouble sleeping, and 4.5% reported always experiencing sleep difficulties. Only 10.4% stated that they never face sleep problems. Since sleep quality is closely related to emotional well-being and mental health, these findings suggest that a considerable number of participants may experience psychological strain consistent with depressive symptoms.

5- How often do you have trouble sleeping (such as trouble falling asleep or staying asleep)? (1 is never, 5 is always)

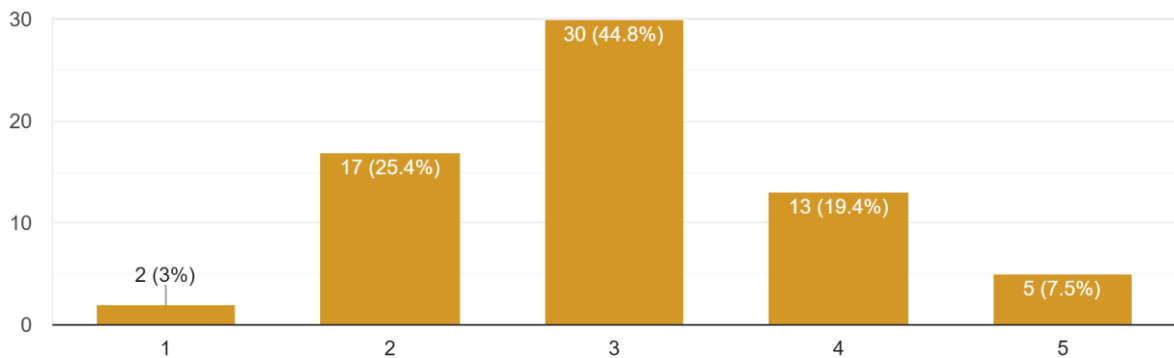
67 responses



In the sixth question, participants were asked how often they feel low in energy or unmotivated during the day. The majority (44.8%) reported experiencing this sometimes, while 25.4% reported rarely feeling so. Furthermore, 19.4% of respondents usually feel low in energy or unmotivated, and 7.5% reported always experiencing this. Only 3% stated that they never feel unmotivated. Since low energy and demotivation are commonly associated with depressive symptoms, these results suggest that a considerable proportion of the sample may experience emotional or psychological strain.

6- How often do you feel low in energy or unmotivated during the day? (1 is never, 5 is always)

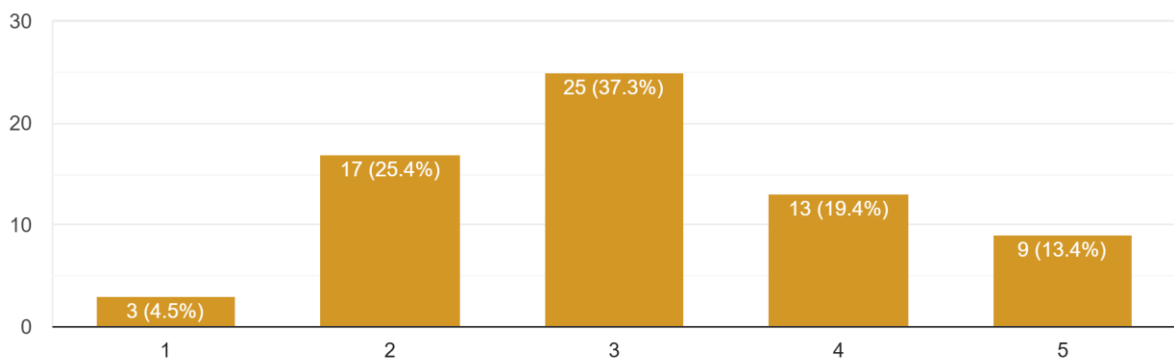
67 responses



In the seventh question, participants were asked how often they prefer spending time alone instead of interacting with others. The largest group (37.3%) reported sometimes preferring solitude over social interaction, while 25.4% reported rarely doing so. Additionally, 19.4% usually prefer to spend time alone, and 13.4% always do so. Only 4.5% stated that they never prefer solitude. Since social withdrawal is often associated with depressive symptoms, these results may indicate the presence of emotional strain in a portion of the sample.

7- How often do you prefer spending time alone instead of interacting with others? (1 is never, 5 is usually)

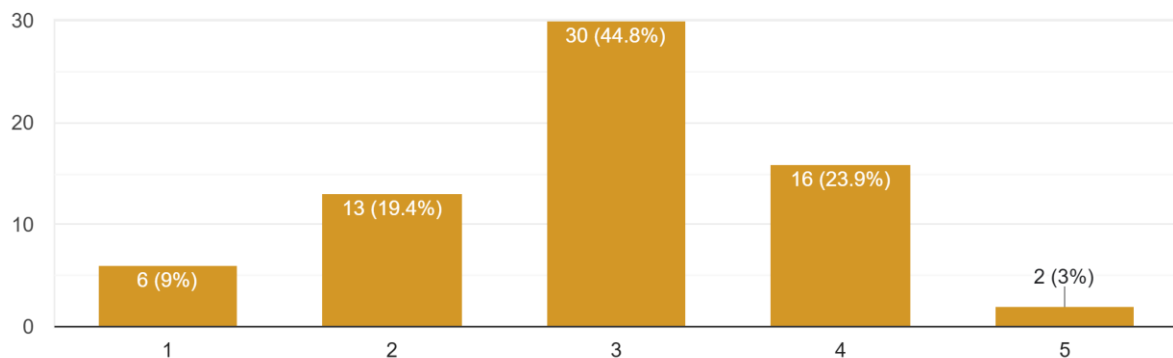
67 responses



In the last question, participants were asked how often they feel overwhelmed when facing stressful situations. The largest group (44.8%) reported feeling overwhelmed sometimes, while 23.9% reported usually feeling overwhelmed. A further 19.4% feel overwhelmed rarely, and 3% reported always feeling overwhelmed. Only 9% stated that they never feel overwhelmed. Since

frequent feelings of overwhelm are commonly associated with anxiety and poor mental well-being, these results suggest the presence of notable psychological strain within the sample.

8- How often do you feel overwhelmed when facing stressful situations? (1 is never, 5 is always)
67 responses



IV. Conclusion

Based on the findings, **the hypothesis that playing video games causes depression is rejected.** The data indicates that the majority of respondents demonstrated moderate rather than extreme levels of depression indicators across all measured dimensions, including sleep quality, energy levels, motivation, and emotional overwhelm. This suggests that video game usage alone does not appear to be a primary driver of depression among the surveyed individuals.

Furthermore, the findings revealed that most respondents did not engage in prolonged daily gaming, and a considerable portion reported using digital entertainment as a means of recharging rather than as a compulsive or harmful habit. Notably, some individuals who reported higher gaming time showed relatively low stress and overwhelm levels, while others with minimal gaming engagement frequently reported feeling unmotivated and emotionally strained. This inconsistency further weakens the proposed causal relationship between gaming and depression.

It is recommended that future studies explore other contributing factors to depression among individuals, such as academic pressure, work-related stress, or social isolation, through a more targeted questionnaire. A suggested hypothesis for future investigation is: academic load causes depression among individuals.

References

ChatGPT, OpenAI (May 2026) <https://chatgpt.com/share/69c82a00-a0f4-832c-baf5-1cda63089dd8>

Appendix

Questionnaire

Old Version

Introduction

We are a group of Computer Science students at the German International University. We are conducting a study to explore various factors related to undergraduates' lifestyles, daily habits, and emotional well-being. The questionnaire is filled anonymously in less than 4 minutes. Your participation is highly appreciated!

Questions

1- Do you experience sleep problems? (General)

- Yes
- No

2- In your free time, what are you most likely to do? (Specific)

- Play video games
- Hang out (Instructor's feedback: with whom? Specify more, use formal language)
- Read a book
- Use social media

(Instructor's feedback: Cornering, must include none of the above)

3- How many hours do you play video games daily? (Specific) (Choose only one please)

(Instructor's feedback: avoid assumption-based questions, revealing; use ODs)

- 0-1 hours (Instructor's feedback: how would that help you to know if they play frequently or not)
- 1-2 hours
- 2-4 hours
- 4+ hours

(Instructor's feedback: cornering, inconsistent choices)

4- On a typical day, how would you describe the overall effect of video games have on your mental state? (General) (Choose One)

(Instructor's feedback: avoid assumption-based questions, ask about 1 variable only in the head of the question, wrong relationship; leads to different hypothesis, revealing hypothesis, must have 5-7 choices)

- They consistently improve my mood and provide energy
- They provide a neutral distraction from daily stress
- They occasionally lead to feelings of frustration or guilt

-They feel like a necessary routine, regardless of how they make me feel

(Instructor's feedback: Cornering)

5- On a scale from 1 to 5, how do you describe your relationship with others? (General)

(Instructor's feedback: Poor navigational instructions, poor OD, missing definition on scale)

1 means Terrible Relations (You see others as enemies and hate being around people)

2 means You face a lot of problems (You feel that the relations you have bring problems to you)

3 means Mild relations (You have good relations, but you see that you are better alone)

4 means Good (You love your current relations, but you see that they need some improvements)

5 means Excellent (You are 100% sure about the people around them and have deep relations with others)

1-----2-----3-----4-----5

6- Which of the following best describes your social experience while gaming during difficult periods? (Choose one answer)

(Instructor's feedback: ask about 1 variable only, wrong relationship leads to wrong hypothesis)

- I seek multiplayer games to feel connected to others

- I prefer single-player games to escape pressure and be alone

- I play games to be near people without having to actively interact

- I find myself withdrawing from gaming entirely when my mood is low

(Instructor's feedback: Cornering)

7- When you feel overwhelmed or stressed, what do you usually do? (Checkbox)

(Instructor's feedback: Poor navigational instructions, inconsistency in choices and their numbers)

- Use my phone

- Sleep or rest

- Go out or change my environment
- Play video games
- Talk to Someone
- Ignore the feeling and continue working

8- On a scale from 1 (Not at all) to 5 (Extremely), to what extent do you use video games as a primary tool to regulate your mood when feeling low or overwhelmed?

(Instructor's feedback: Poor navigational instructions, wrong relationship, ask about 1 variable in the question)

1-----2-----3-----4-----5

Demographics

Age:

Gender: -Male -Female

Major/Profession:

Updated version

Introduction

We are a group of Computer Science students at the German International University. We are conducting a study to explore various factors related to undergraduates' lifestyles, daily habits, and emotional well-being. The questionnaire is filled anonymously in less than 4 minutes. Your participation is highly appreciated!

Questions

1- Which of the following describes how you recharge after a long day? (Check all that apply)

a- Social Activities

- b- Solo Activities
- c- Digital Entertainment
- d- Outdoor Activities
- e- Creative hobbies
- f- Sleep
- g- None of the above

2- Do you find it difficult to stay calm in stressful situations? (Choose one answer)

- Yes
- No

3- In your free time, what are you most likely to do? (Choose one answer)

- a- Hang out with a friend
- b- Play digital Interactive games on my PC)
- c- Read a book
- d- Use Social Media
- e- Develop a skill
- f- None of the above

4- On a typical day, how much time do you spend gaming on your phone? (Choose one answer)

- a- Less than 1 hour
- b- 1–2 hours
- c- 2–3 hours
- d- More than 3 hours
- e- None of the above

5- How often do you have trouble sleeping (such as trouble falling asleep or staying asleep)? (1 is never, 5 is always)

1-----2-----3-----4-----5

6- How often do you feel low in energy or unmotivated during the day?

(1 is never, 5 is always)

1-----2-----3-----4-----5

7- How often do you prefer spending time alone instead of interacting with others? (1 is never, 5 is usually)

1-----2-----3-----4-----5

8- How often do you feel overwhelmed when facing stressful situations?

(1 is never, 5 is always)

1-----2-----3-----4-----5

Demographics

Age:

Gender: -Male -Female

Major/Profession: